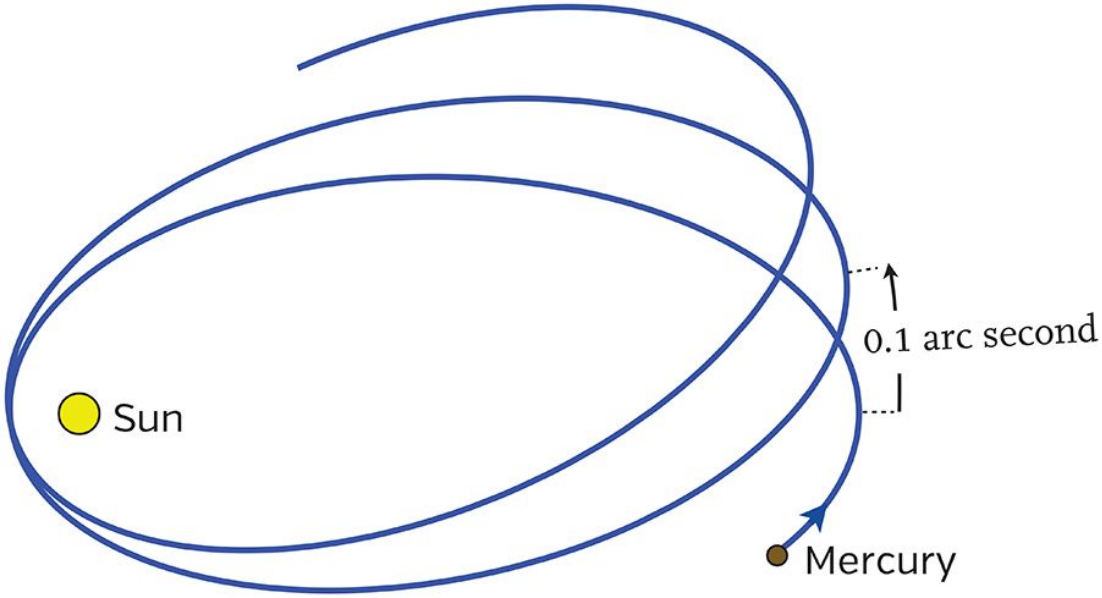


Fig. 24.1. The anomalous precession of Mercury's orbit. In this picture, I exaggerate the orbit's ellipticity (its elongated shape) and the magnitude of its precession.

Today the measured anomalous precession and the predictions by Einstein's laws agree to within one part in a thousand (one-thousandth of the anomalous precession), which is the accuracy of the observations —a great triumph for Einstein!

The Anomalous Orbits of Galaxies Around Each Other

In 1933 the Caltech astrophysicist Fritz Zwicky announced he had discovered a huge anomaly in the orbits of galaxies around each other. The galaxies were in the Coma cluster (Figure 24.2), a collection of about a thousand galaxies, 300 million light-years from Earth, in the constellation Coma Berenices.

From the Doppler shifts of the galaxies' spectral lines, Zwicky could estimate how fast they were moving relative to each other. And from the brightness of each galaxy, he could estimate its mass and thence its